

Understanding Migration Responses to Local Shocks

Borusyak, Dix-Carneiro & Kovak 2022

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Motivation

Economists studying migration often rely on the **conventional migration regression** (CMR):

$$\hat{L}_l = \alpha + \beta \hat{z}_l + \epsilon_l$$

That is, the population change in location ℓ is regressed on the observed shock to that same location.

- Implicitly, this argues that the migration response is locally determined
- β is commonly found to be at or near zero

Empirical puzzle

Could it be that migration responds little to local shocks and plays no role in adaptation?

That's not the full story

The paper's central point is that migration is a bilateral choice shaped by pre existing networks.

If so, the CMR omits:

- shocks to destinations and other outside options;
- the pre-existing origin–destination migration network;
- industry-switching frictions when shocks are industry-based.

⇒ A small β may reflect **correlated shocks across migrant-connected places**, not low mobility.

This paper

Borusyak et al argue the CMR is misspecified. To demonstrate that, they:

- ① Build a simple model of regional labor markets with mobility costs.
- ② Solve for the population response to a vector of local shocks.
- ③ Characterize the CMR estimand β under this data-generating process.
- ④ Propose model-consistent alternatives
- ⑤ assess them in simulations using Brazilian data.

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Setup

We focus on the baseline model with free mobility across industries.

Environment

- sectors $n \in \mathcal{N}$, locations $\ell \in \mathcal{L}$;
- products differentiated by sector and production location;
- no trade costs; labor is the only input;
- perfect competition: $p_{\ell n} = w_{\ell} / \varphi_{\ell n}$.

Workers

$$V_{iol} = \ln\left(\frac{w_{\ell}}{P}\right) - \ln \tau_{ol} + \frac{1}{\theta} \epsilon_{il},$$

where:

P is the Cobb Douglas price index

$\ln(\tau_{ol})$ is a cost of moving

ϵ_{il} is type-I extreme value

θ is location preference strenght

Extensions allow industry-switching costs, inelastic housing supply, local agglomeration economies **and exogenous ammenities**.

Labor Demand

Perfect competition and regional trade balance imply that local labor demand can be summarized by a regional demand shifter.

$$L_\ell = D_\ell w_\ell^{-\sigma},$$
$$D_\ell \equiv \sum_n \eta_n (\varphi_{\ell n} P_n)^{\sigma-1}.$$

Let hats denote small proportional changes: $\hat{x} \equiv dx/x = d \ln x$.

$$\hat{L}_\ell = \hat{D}_\ell - \sigma \hat{w}_\ell, \quad \hat{L} = \hat{D} - \sigma \hat{w}.$$

σ is the labor demand elasticity

Labor Supply

Workers from o choose destination ℓ with probabilities $\pi_{o\ell}$ induced by the extreme-value taste shocks.

$$\pi_{o\ell} = \frac{(w_\ell/\tau_{o\ell})^\theta}{\sum_d (w_d/\tau_{od})^\theta}, \quad L_\ell = \sum_o \pi_{o\ell} L_o^0.$$

Let the count of workers from o living in ℓ be $f_{o\ell} \equiv \pi_{o\ell} L_o^0$ and the in-migration share

$$\gamma_{o\ell} \equiv \frac{f_{o\ell}}{L_\ell}.$$

Taking the total derivative with respect to wages gives, in matricial notation

$$\hat{\mathbf{L}} = \theta (I - \Gamma' \Pi) \hat{\mathbf{w}}.$$

θ governs responsiveness to real-wage differences; Γ and Π encode the migration network.

Equilibrium

Equating supply and demand yields the equilibrium response of population to regional demand shocks:

$$\hat{L} = \Omega \hat{D},$$

$$\Omega \equiv I - \left(I + \frac{\theta}{\sigma} (\mathbf{I} - \Gamma' \Pi) \right)^{-1}.$$

What modulates the local response?

The effect of $\hat{D}_\ell$ depends on θ/σ , the migration network $\Gamma'\Pi$, and the shocks to all other locations.

If all locations receive the same shock, relative location values are unchanged and population is unchanged.

Heterogeneity is of first order relevance

Under a low-mobility approximation, the population response becomes intuitive:

$$\hat{L}_\ell \approx \frac{\theta}{\sigma} \left(\sum_o \gamma_{o\ell} (\hat{D}_\ell - \hat{D}_o) + \sum_d \pi_{\ell d} (\hat{D}_\ell - \hat{D}_d) \right).$$

Interpretation

A shock to location ℓ matters through comparisons with shocks to migrant-connected places:

- origins that typically send migrants to ℓ receive weight $\gamma_{o\ell}$;
- destinations that people from ℓ typically choose receive weight $\pi_{\ell d}$.

Core message: population responses inherit heterogeneity from the migration network.

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CMR β under the Model

Let $\hat{z}_\ell$ be the observed shock to location ℓ . The CMR is

$$\hat{L}_l = \alpha + \beta \hat{z}_l + \epsilon_l$$

Three definitions will be needed:

- ① Total number of migrants: $M \equiv \sum_{o \neq d} f_{od}$ (M/L = migrant share of the population)
- ② Average shock correlation weighted by migration flows $\rho = \sum_{o \neq d} \frac{f_{od}}{M} \text{Corr}[\hat{z}_o, \hat{z}_d]$
- ③ No-mobility counterpart to variable x , $\tilde{x}$

Theorem 1: CMR β under the model

$$\beta = \frac{2\theta}{\sigma} \cdot \frac{M/L}{\widetilde{M}/L} \cdot \frac{1 - \rho}{1 - \tilde{\rho}}.$$

Issues with the CMR β

The true causal effect of a shock to one location is not the same object as the CMR coefficient!

True model logic

- compares ℓ to its migration-weighted outside options;
- varies with local migration intensity M_ℓ/L_ℓ ;
- responds to the full vector of shocks.

CMR logic

- compares $\hat{z}_\ell$ to the mean shock;
- embeds only national migration intensity;
- is attenuated when connected regions face similar shocks.

A Model Consistent Alternative

If Π and Γ are known, the model gives a direct estimation target for NLLS.

$$\hat{\mathbf{L}} = \Omega(\lambda)\hat{\mathbf{z}} + \zeta, \quad \Omega(\lambda) = \mathbf{I} - (\mathbf{I} + \lambda(\mathbf{I} - \Gamma'\Pi))^{-1}, \quad \lambda \equiv \frac{\theta}{\sigma}.$$

$$\min_{\lambda} \mathbb{E} \left[\left(\hat{\mathbf{L}} - \Omega(\lambda)\hat{\mathbf{z}} \right)' \left(\hat{\mathbf{L}} - \Omega(\lambda)\hat{\mathbf{z}} \right) \right].$$

Lower-information alternatives

Under low mobility, a model-consistent linear specification is viable without knowing Π and Γ . At the pair level,

$$\hat{f}_{ol} = \alpha + \beta_1 \hat{z}_\ell + \beta_2 \hat{z}_o + \varepsilon_{ol}, \quad \beta_1 = -\beta_2 = \frac{\theta}{\sigma}.$$

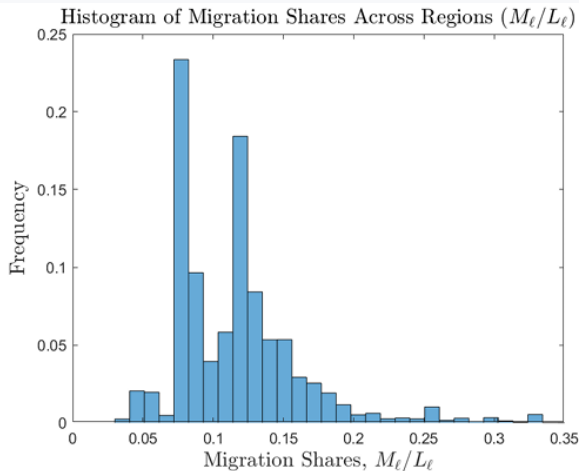
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Heterogeneity in Data

The quantitative exercise uses 1999-2010 RAIS data to recover the migration network matrices.

Mobility is far from uniform:
the local migration share M_ℓ/L_ℓ varies substantially across regions.

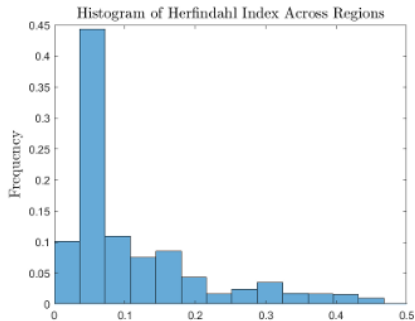


Networks in Data

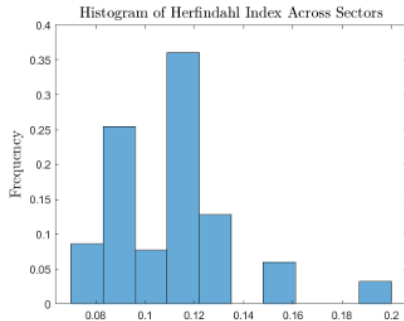
Migration and industry switching are networked: flows concentrate in a limited set of connected locations and industries.

- Each location is connected to 9 others out of 485 possible ones (9 to 24 possible in industries)

Panel (a): Across Locations



Panel (b): Across Industries



Calibration and Benchmark Prediction

How well do different procedures predict the population response when shocks are fully known?

Calibration and Simulation

- $\theta = 1.63$ and $\sigma = 4$ from literature estimates.
- Π and Γ from average RAIS flows.
- 500 simulations ran

Prediction comparison: Mean MSE across simulations

$$\text{MSE} = \frac{1}{S} \sum_{s=1}^S \left[\sum_{\ell} \frac{L_{\ell}}{L} \left(\widehat{L}_{\ell;s}^{\text{true}} - \widehat{L}_{\ell;s}^{\text{predicted}} \right)^2 \right].$$

Benchmark: Model MSE relative to a no-change prediction ($\widehat{L}_{\ell;s}^{\text{predicted}} = 0$)

Assessment

Prediction method	MSE relative to $\widehat{L}_\ell = 0$	
	iid location shocks	iid industry shocks
Conventional regression	0.244	0.514
NLLS, region-industry level	0.026	0.001
NLLS, misspecified regional model	0.033	0.358
OLS, low-mobility approximation	0.049	0.177
Distance-weighted average of $\widehat{z}_k$	0.229	0.666

Reading the table

NLLS using the correctly specified region-industry model nearly eliminates prediction error. The conventional regression performs poorly, especially for industry shocks.

Migration spillovers matter; industry mobility is crucial when shocks are industry-driven.

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What we've learned

Main lesson: a small CMR coefficient need not mean that migration is unresponsive to shocks.

- Outcomes to each location depend on the full vector of local shocks.
- Pre-existing migration networks determine which outside options matter.
- Correlated shocks across connected regions attenuate the conventional coefficient.
- Industry shocks add further attenuation when workers face industry-switching costs.
- Model-consistent NLLS is robust in the simulations; low-mobility OLS alternatives can help when data are thinner.

For empirical work

Do not use the CMR coefficient alone to infer mobility parameters or to justify ignoring migration.